MODULE 5

03/Nov/2023

Module - 5

Test of Significance for Small Samples

Student T - Distribution / T - Distribution:

If $x_1, x_2, x_3, \ldots x_n$, be a random sample of size n (<30) drawn from a normal population with mean μ and variance σ^2 then the test statistic 't' is defined by

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

where $\bar{x}$ is sample mean and $\boxed{s^2 = \frac{\Sigma (x_i - \bar{x})^2}{n-1}}$ is an un biased estimation of σ^2. If the test statistic 't' is a random variable having the 't' - Distribution with $(n-1)$ degrees of freedom (d.f) This is known as Student T - Distribution (or) T - Distribution.

Degrees of Freedom: The no. of independent variates which make the test statistic is known as degrees of freedom and it is denoted by 'v'.

T - Test for Single mean:

Suppose we want to test

1. If a random sample (x_i) of size n has been drawn from a normal population with a specified mean μ.
2. If the sample mean different significantly from the hypothecally value μ of the population mean.

Working Rule:

Let a random sample of size n (<30) has a sample mean $\bar{x}$. To test the hypothesis that the population mean μ has the specified value of μ_0, when population standard deviation σ is unknown.

1. Null Hypothesis :- $H_0 : \bar{x} = \mu \,/\, \mu = \mu_0$
2. Alternative Hypothesis :- $H_1 : \mu \neq \mu_0 \,/\, \bar{x} \neq \mu$
3. Level of Significance :- α

4. Test Statistic : $t = \frac{\bar{x}-\mu}{s/\sqrt{n-1}}$

where 's' is a sample standard deviation follows t-Distribution with $v = n-1$ degrees of freedom.

1. A Sample of 26 Bulbs gives a mean life of 990 hours with a S.D of 20 hrs. The manufacturer claims the mean life of bulbs is 1000 hrs. Is the sample not up to the standard.

Sol: Given,

$n = 26$

$\bar{x} = 990$

$s = 20$

$\mu = 1000$

1. Null Hypothesis :- H_0 : The sample up to a standard $(\bar{x} = \mu)$

2. Alternative Hypothesis :- H_1 : The sample not upto a standard $(\bar{x} \neq \mu)$

3. Level of Significance :- $\alpha = 0.05 / 5\% / 95\%$ (Say)

4. Test statistic :- $t = \frac{\bar{x}-\mu}{s/\sqrt{n-1}} = \frac{990-1000}{20/\sqrt{25}} = -2.5$

$t = -2.5$

$\therefore \boxed{|t| = 2.5}$

5. Decision :- The calculated value of $|t|$ is 2.5 and the critical value of $t_{\frac{\alpha}{2}}$ in 0.05 at 25 of is 1.708

$\therefore$ Hence $|t| > t_{\frac{\alpha}{2}}$, we conclude that, the Null Hypothesis is rejected.

Student - T - Test (When standard Deviation of the sample is not given directly) :

1. A random sample of 10 Boys have the following IQ's 70, 120, 110, 101, 88, 83, 95, 98, 107 and 100. Do these data suppose the assumption of the population mean IQ of 100

Sol:-

Given,

$n = 10$

$\bar{x} = \frac{70+120+\cdots 107+100}{10}$

$\boxed{\bar{x} = 97.2}$

$s^2 = \frac{\Sigma(x_i-\bar{x})^2}{n-1} = \frac{(70-97.2)^2+(120-97.2)^2+\cdots+(100-97.2)^2}{10-1}$

$s = 14.27$

1. Null Hypothesis :- $H_0 : \mu = 100$
2. Alternative Hypothesis :- $H_1 : \mu \neq 100$
3. Level of significance : $\alpha = 5\%$ (say)
4. Degrees of freedom (D.f) : $v = n-1 = 9$
5. Test statistic : $t = \frac{\bar{x}-\mu}{s/\sqrt{n-1}} = \frac{97.2-100}{14.27/\sqrt{9}} = -0.6$

$\boxed{|t| = 0.6}$

Decision :- The calculated value of $|t|$ is 0.6 and the critical value of $t_{\frac{\alpha}{2}}$ in 0.05 at 9 of is 1.833

Hence $|t| < t_{\frac{\alpha}{2}}$, we conclude that the Null Hypothesis is Accepted.

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Pair Sample T-Test / Two-sample T-Test :-

If $(x_1, y_1), (x_2, y_2), \ldots\ldots (x_n, y_n)$, be the pairs of sets data before and after the sales promotions in a business concern, we apply pair T-test to examine the significance of the difference of two situations.

Let $d_i = x_i - y_i$ / $y_i - x_i$ for $i = 0, 1, 2$

Procedure :

1. Null Hypothesis : $H_0 : \mu_1 = \mu_2$ (i.e $\mu = 0$) (There is no Significance b/w two situations)
2. Alternative Hypothesis : $H_1 : \mu_1 \neq \mu_2$
3. Level of Significance : α

(4). Degrees of freedom (D.f): $v = n-1$

(5). Test statistic:

$$\boxed{t = \frac{\bar{d} - \mu}{s/\sqrt{n}}} \quad (\text{ie } \mu = 0)$$

where $\bar{d} = \frac{\Sigma d_i}{n}$

$$s^2 = \frac{\Sigma (d_i - \bar{d})^2}{n-1}$$

The above test statistic follows students T-Test with n-1 degrees of freedom.

1. The blood pressures of 5 women before and after intake of a certain drug are given below

Before	110	120	125	132	125
After	120	118	125	136	121

, Test whether there is significant change in blood pressure at 1% L.O.S

Sol: Given,

$n = 5$

1. Null Hypothesis: $H_0: \mu_1 = \mu_2$ (There is no significance before and after)
2. Alternative Hypothesis: $H_1: \mu_1 \neq \mu_2$ (There is significance before and after)
3. Level of Significance: $\alpha = 0.01 / 1\%$
4. Degrees of freedom: $v = n-1 = 4$, $\bar{d} = \frac{\Sigma d_i}{n} = \frac{8}{5} = 1.6$

Before (x_i)	After (y_i)	$d_i = y_i - x_i$	$d_i - \bar{d} = d_i - 1.6$	$(d_i - \bar{d})^2$
110	120	10	8.4	70.56
120	118	-2	-3.6	12.96
125	125	0	-1.6	2.56
132	136	4	2.4	5.76
125	121	-4	-5.6	31.36
		$\Sigma d_i = 8$		$\Sigma (d_i - \bar{d})^2 = 123.2$

$$s^2 = \frac{\Sigma(d_i - \bar{d})^2}{n-1}$$

$$s^2 = \frac{123.2}{4}$$

$$s^2 = 30.8$$

$$\boxed{s = 5.5}$$

(v) The Test statistic is

$$t = \frac{\bar{d} - \mu}{s/\sqrt{n}}$$

$$t = \frac{1.6 - 0}{5.5/\sqrt{5}} = 0.6$$

Decision:

The calculated value t is 0.6 and the critical value of t_α in 4 d.f at 1% LOS is 3.747

Hence $|t| < t_\alpha$ then we conclude that the Null Hypothesis is Accepted.

Thus, there is no significance before and after.

F-Test of Significance for Equality of two standard deviation

Let two independent random samples of sizes n_1 and n_2 be drawn from two populations.

Working Rule:

To test the hypothesis the two standard deviation σ_1 and σ_2 are equal

1. Null Hypothesis: $H_0 : \sigma_1 = \sigma_2$
2. Alternative Hypothesis: $H_1 : \sigma_1 \neq \sigma_2$
3. Level of Significance: α
4. Degrees of freedom: $v = (v_1, v_2)$
   $v_1 = n_1 - 1, v_2 = n_2 - 1$
5. Test statistic:

$$F = \frac{s_1^2}{s_2^2}, \text{ if } s_1^2 > s_2^2$$

(or)

$$F = \frac{s_2^2}{s_1^2}, \text{ if } s_2^2 > s_1^2$$

where,

$$s_1^2 = \frac{\Sigma(x_i-\bar{x})^2}{n_1-1}$$

$$s_2^2 = \frac{\Sigma(y_i-\bar{y})^2}{n_2-1}$$

1. The time taken by workers in performing a Job by method 1 and method 2 is given below

method I	20	16	26	27	23	22	–
method II	27	33	42	35	32	34	38

Do the data show that the variance of time distribution from population for which these samples are drawn do not diffence significantly.

Ans

1. Null Hypothesis : $H_0 : \sigma_1 = \sigma_2$
2. Alternative Hypothesis : $H_1 : \sigma_1 \neq \sigma_2$
3. Level of significance : $\alpha = 5\%$ (Say)
4. Degrees of freedom : $V = (n_1-1, n_2-1)$

$V = (5, 6)$

$\bar{x} = \frac{\Sigma x_i}{n_1} = \frac{134}{6} = 22.3$

$\bar{y} = \frac{\Sigma y_i}{n_2} = \frac{241}{7} = 34.7$

method I (x_i)	$x_i - \bar{x}$	$(x_i-\bar{x})^2$	method II y_i	$y_i-\bar{y}$	$(y_i-\bar{y})^2$
20	−2.3	5.29	27	−7.4	54.76
16	−6.3	39.69	33	−1.4	1.96
26	3.7	13.69	42	7.6	57.76
27	4.7	22.09	35	0.6	0.36
23	0.7	0.49	32	−2.4	5.76
22	−0.3	0.09	34	−0.4	0.16
–	–	–	38	3.6	12.96
134		81.34	241		133.72

Now, $s_1^2 = \frac{\Sigma(x_i-\bar{x})^2}{n_1-1} \Rightarrow s_1^2 = \frac{81.34}{5}$

$\therefore \boxed{s_1^2 = 16.26}$

$$S_2^2 = \frac{\Sigma (y_i - \bar{y})^2}{n_2 - 1}$$

$$= \frac{133.72}{6} = 22.29$$

$\therefore S_2^2 > S_1^2$

Then, the test statistic

$$F = \frac{S_2^2}{S_1^2} = \frac{22.29}{16.26} = 1.37$$

Decision:

The calculated value of F is 1.37 and the Critical value of F_α in (5,6) d.f at 5% L.O.S is 4.39

Hence $|F| < F_\alpha$, we conclude that the Null Hypothesis is Accepted.

08/Nov/2023

CHI-SQUARE Test for Independence of Attributes:-

If a set of events $A_1, A_2, A_3, \ldots A_n$ are observed to occur with frequencies $O_1, O_2, O_3, \ldots O_n$ respectively and According to probability rules $A_1, A_2, \ldots A_n$ are expected to occur with frequencies $E_1, E_2, E_3 \ldots E_n$ respectively with $O_1, O_2, O_3, \ldots O_n$, are called observed frequencies. Then the χ^2 is defined as

$$\boxed{\chi^2 = \sum_{i=1}^{n} \left[\frac{(O_i - E_i)^2}{E_i} \right]}$$

with n-1 degrees of freedom

Definition:

Literally an attribute means a quality or characteristic

Example: Blindness, Beauty, Drinking, Smoking, etc

Working Rule:

1. **Null Hypothesis:** Ho : There is no association b/w the attributes (or) The two attributes are independence
2. **Degrees of freedom:** $v = (\text{No. of rows} - 1) \times (\text{No. of columns} - 1)$
3. **Level of significance:** α

(4). The Expected frequencies (Ei) of any cell $= \frac{\text{row total} \times \text{column total}}{\text{Grand total}}$

(5) The test statistic:

$$\chi^2 = \sum_{i=1}^{n} \left(\frac{(O_i - E_i)^2}{E_i} \right)$$

(6) Conclusion:- If the calculated value of χ^2 is less than the tabulated value / critical value of a specified L.O.S α then the Null Hypothesis holds good. On other hand, if the calculated value of χ^2 is greater than the specified value of L.O.S of α then we say that the attributes are associated.

The various cell-frequencies can be expressed in the following table known as 2×2 contigency table

A	a	b
B	c	d

The Expected Frequencies are given by

a	b	a+b
c	d	c+d
a+c	b+d	N=a+b+c+d

$E(a) = \frac{(a+c)\times(a+b)}{N}$	$E(b) = \frac{(b+d)\times(a+b)}{N}$	a+b
$E(c) = \frac{(a+c)\times(c+d)}{N}$	$E(d) = \frac{(b+d)\times(c+d)}{N}$	c+d
a+c	b+d	N=a+b+c+d

1. From the following data find whether there is any significant liking in the habit of taking soft drink among the categories of employees

Soft Drinks	clerks	Teachers	Officers
Pepsi	10	25	65
Thumsup	15	30	65
Fanta	50	60	30

Sol:-

1. Null Hypothesis :- H_0: There is a significant liking in the habit of taking soft drinking.

2. Alternative Hypothesis: H_1:- There is no significant liking in the habit of taking soft drinking.

3. Degrees of freedom :- $\nu = (\text{No. of rows} - 1) \times (\text{No. of columns} - 1)$
$= 4$

4. Level of Significance:- $\alpha = 5\%$ (Say)

5. The Expected frequencies (E_i) of any cell

$$= \frac{\text{row total} \times \text{column total}}{\text{Grand total}}$$

6. The test statistic:

$$\chi^2 = \sum_{i=1}^{n} \left(\frac{(O_i - E_i)^2}{E_i} \right)$$

Computations:-

Soft Drinks	clerks	Teachers	Officers	Total
Pepsi	10	25	65	100
Thumbup	15	30	65	110
Fanta	50	60	30	140
Total	75	115	160	350

$\frac{75 \times 100}{350} = 21.43$	$\frac{100 \times 115}{350} = 32.86$	$\frac{100 \times 160}{350} = 45.71$	100
$\frac{75 \times 110}{350} = 23.57$	$\frac{110 \times 115}{350} = 36.14$	$\frac{110 \times 160}{350} = 50.29$	110
$\frac{75 \times 140}{350} = 30$	$\frac{115 \times 140}{350} = 46$	$\frac{140 \times 160}{350} = 64$	140
75	115	160	350

calculations of χ^2

Observed frequencies (O_i)	Expected frequencies (E_i)	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
10	21.43	-11.4	129.96	6.07
25	32.9	-7.9	62.41	1.90
65	45.7	19.3	372.49	8.15
15	23.6	-8.6	73.96	3.13
30	36.1	-6.1	37.21	1.03
65	50.3	14.7	216.097	4.30
50	30	20	400	13.33
60	46	14	196	4.26
30	64	34	1156	18.06
				= 60.23

$$\chi^2 = \sum_{i=1}^{n} \left(\frac{(O_i - E_i)^2}{E_i} \right)$$

$$\therefore \boxed{\chi^2 = 60.23}$$

Decision:

The calculated value of χ^2 is 60.23 and the critical value of α at 5% L.O.S is 9.488

Hence $\chi^2 > \chi^2_\alpha$, we conclude that the Null Hypothesis is Rejected.

The End